

WASP-52b

R-1850

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_220807 · created 2026-08-03T08:09:18+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459798.9303 +/- 0.0068 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.215 +/- 0.06
Transit depth (Rp/Rs)^2	4.64 +/- 2.59 [%]
Orbital Inclination (inc)	84.33 +/- 2.49
Airmass coefficient 1 (a1)	0.181 +/- 0.061
Airmass coefficient 2 (a2)	1.45 +/- 0.64
Scatter in the residuals of the lightcurve fit is	30.52 %
Best Comparison Star	#3 - [126, 309]
Optimal Aperture	0.75
Optimal Annulus	4.75
Transit Duration (day)	0.067 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.215 ± 0.06 vs 0.1646 (deviation 0.42σ)
Duration fit vs published	0.067 d vs 0.0758 d (deviation 0.18σ)
Mid-time O–C	-0.5 min
Depth SNR	1.8
Residual scatter	30.52 %

Light curve

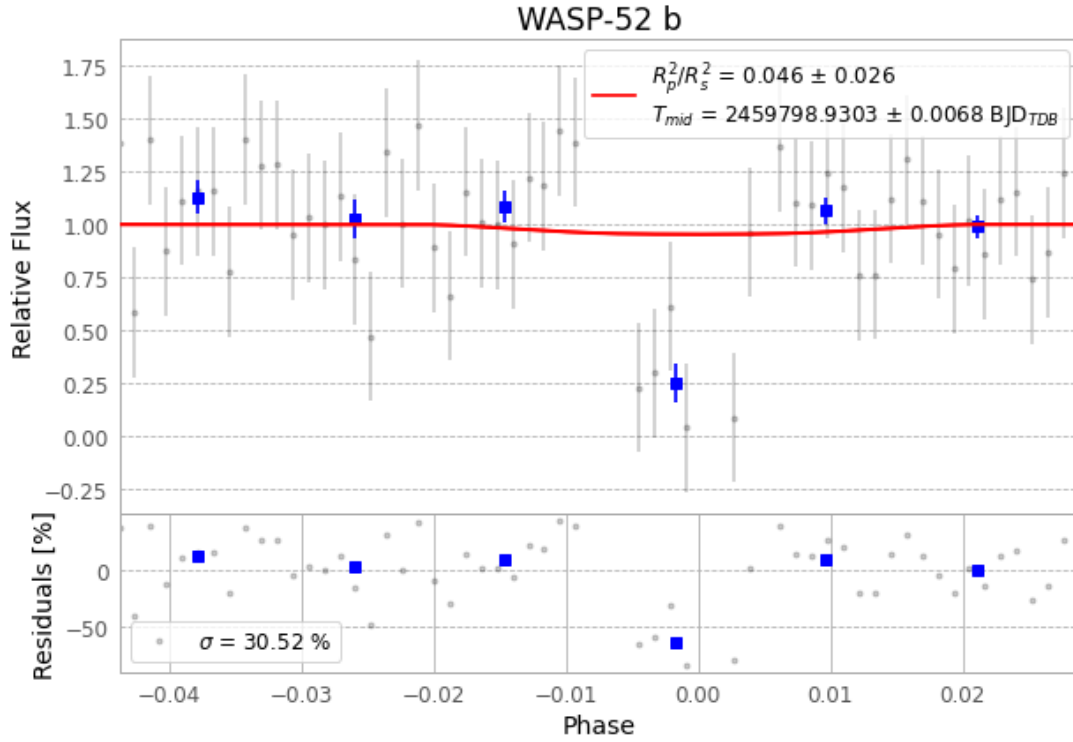


Figure 1 — FinalLightCurve_WASP-52 b_2022-08-07.png (SHA-256 f555cbaa450bce4b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [636, 84], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 50410ae9e45c1240...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

62 FITS frames (41 MB), WASP-52220807082113.FITS → WASP-52220807112413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-220807.log (e7a889f6620f...), FinalLightCurve_WASP-52 b_2022-08-07.png (f555cbaa450b...), FinalLightCurve_WASP-52 b_2022-08-07.csv (6b14950194ce...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 08:09Z.